Full Stack Assignment – 01

Assignment ID: FSD-01

Title: Resource Allocation Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Resource Allocation Management System** that allows users to request resources and administrators to manage allocations.
The system must enforce allocation rules and ensure **data persistence using a database**.

2️⃣ User Roles

USER

- Register and log in to the system
- View available resources
- Request allocation of a resource
- View allocation status

ADMIN

- Log in to the system
- View all allocation requests
- Approve or reject resource allocation requests
- View current resource allocation status

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all APIs using JWT
- Enforce **role-based access control** on both backend APIs and frontend views

Resource Management

- Create and manage resources (ADMIN only)
- Store all resource data in the database
- Fetch resource availability dynamically from the database

Allocation Requests

- USER can request a resource

- ADMIN can approve or reject requests
- Allocation status must be stored and retrieved from the database

4️⃣ Business Rule (Critical)

⚠️ **A resource can be allocated to only one user at a time.**

If a resource is already allocated, further allocation requests for the same resource must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All data **must persist in a database**
- Hardcoded or in-memory data storage is **not allowed**
- Backend APIs must reflect real database state
- JWT authentication and role checks are mandatory
- Environment variables must be used for sensitive data

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Display list of available resources
- Show allocation status clearly
- Allow ADMIN to approve/reject requests

UI design is not evaluated; functionality and clarity are the priority.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview

- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links